before we looked.[1] Thus, if we want to replicate an object, we face a quantum Catch-22. To replicate we must observe, so we know what to replicate. But the act of observation causes change, so if we replicate what we see, we will not replicate what was there before we looked. This suggests that teleportation in a quantum universe is unattainable, not merely because of practical limitations arising from complexity, but because of fundamental limitations inherent in quantum physics. Nevertheless, as we'll see in the next section, in the early 1990s an international team of physicists found an ingenious way to circumvent this conclusion.

As for the first question, regarding the relationship between replica and original, quantum physics gives an answer that's both precise and encouraging. According to quantum mechanics, every electron in the universe is identical to every other, in that they all have exactly the same mass, exactly the same electric charge, exactly the same weak and strong nuclear force properties, and exactly the same total spin. Moreover, our well-tested quantum mechanical description says that these *exhaust* the attributes that an electron can possess; electrons are all identical with regard to these properties, and there are no other properties to consider. In the same sense, every up-quark is the same as every other, every down-quark is the same as every other, every photon is the same as every other, and so on for all other particle species. As recognized by quantum practitioners many decades ago, particles may be thought of as the smallest possible packets of a field (e.g., photons are the smallest packets of the electromagnetic field), and quantum physics shows that such smallest constituents of the same field are always identical. (Or, in the framework of string theory, particles of the same species have identical properties because they are identical vibrations of a single species of string.)

What can differ between two particles of the same species are the probabilities that they are located at various positions, the probabilities that their spins are pointing in particular directions, and the probabilities that they have particular velocities and energies. Or, as physicists say more succinctly, the two particles can be in different *quantum states.* But if two particles of the same species are in the same quantum state—except, possibly, for one particle having a high likelihood of being *here* while the other particle has a high likelihood of being over *there*—the